

1. (Fall 2024, #6) Using appropriate tests, determine whether the series below converge or diverge. Show all work, by naming the test, applying the test, and drawing the correct conclusion.

(a) $\sum_{n=1}^{\infty} \frac{2^n}{n + 5^n}$

(b) $\sum_{n=3}^{\infty} \frac{1}{\sqrt[3]{4n^3 + 4n + 1}}$

(c) $\sum_{n=1}^{\infty} (n^2 + 1) \sin \frac{1}{n^2}$

2. (Spring 2024, #7) Determine the convergence of each series. Circle your answer in each case and justify it carefully, clearly stating any tests you use.

(a) $\sum_{n=0}^{\infty} (\sin n) \left(\frac{n+3}{5n+7} \right)^{2n}$ Absolutely convergent Conditionally convergent Divergent

(b) $\sum_{n=0}^{\infty} \frac{1}{n(\ln n)^3}$ Convergent Divergent

3. (Fall 2023, #6) Determine whether the following series are conditionally convergent, absolutely convergent, or divergent. In each case, be sure to carefully justify your answer.

(a)
$$\sum_{n=1}^{\infty} \ln \left(\frac{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)}{2 \cdot 5 \cdot 8 \cdot \dots \cdot (3n-1)} \right)$$

(b)
$$\sum_{n=4}^{\infty} \frac{(-1)^n \cdot n^{3/2}}{n^3 + n^2 \cos(n)}$$

(c)
$$\sum_{n=2}^{\infty} \frac{(-1)^n}{n \sqrt{\ln(n^2)}}$$

4. (Spring 2023, #7) Determine whether the following series are absolutely convergent, conditionally convergent, or divergent. Clearly state any test(s) you use and show that they are applicable. *Circle your answer for each part.*

(a) $\sum_{k=1}^{\infty} \frac{\sin k}{k^{1.26}}$

Absolutely convergent

Conditionally convergent

Divergent

(b) $\sum_{n=1}^{\infty} \left(\frac{1}{n} - 1\right)^n$

Absolutely convergent

Conditionally convergent

Divergent

5. (Fall 2022, #6) Determine whether the following series converge or diverge. Clearly state the method(s) you use.

(a)
$$\sum_{n=1}^{\infty} \frac{n^2 + 2}{\sqrt{n^7} + 1}$$

(b)
$$\sum_{n=2}^{\infty} \frac{\ln n}{n^2}$$

(c)
$$\sum_{n=1}^{\infty} \ln \left(1 + \frac{1}{n} \right)$$